Capacitively-Coupled Chopper Instrumentation Amplifier for Linear Hall Effect Sensor

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Team Chop

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1 Introduction

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- Discuss magnetic sensing application space
  - Current sensing
  - Tactile Sensing (robotics)
  - Position/Velocity Sensing (automotive)
  - Biosensors
- Discuss utility of building open-source Hall sensor readout
  - No existing data on n-well-based Hall sensors in open PDKs
  - Sensor evaluation gives future designers ability to integrate into their application
  - Readout circuit directly applicable to most bridge-type sensors

2 Conceptual Design

- Show high-level block diagram
- Table of pinouts with signal descriptions/directions

2.1 Sensor

- Vertical vs. Horizontal sensor design
- Horizontal sensor geometries
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2.2 Readout Circuit

- Instrumentation Amplifier options/survey
- CCIA chosen due to large common-mode input range (sweepable bias current), stable gain (easy to integrate several gain options if needed)
- Hall sensor bias expected to dominate power consumption, but CCIA being low-power is great in other high-impedance sensor applications
- analog output reduces complexity and allows for easier characterization of CCIA

3 Design Specifications

- Table of specifications for full system

3.1 Sensor

3.2 Readout Circuit

4 Schematic Design

- Show top-level schematic

4.1 Subblock 1

4.2 Subblock N

5 Layout Design

- Show layout floorplan

5.1 Subblock 1

5.2 Subblock N

6 Simulation Results

- Verification Plan details (how is each specification tested?)

6.1 Simulation 1

6.2 Simulation M

References

[1] L. Lamport, *LaTeX: A Document Preparation System*, 2nd ed. Addison-Wesley, 1994.